COMPARATIVE ELECTROCHEMICAL IMPEDANCE SPECTROSCOPY AND CURRENT–VOLTAGE CHARACTERIZATION OF LOW-COST COMMERCIAL SILICON SOLAR CELLS

Kevin Morkar[1],[2] [1]Department of Electrical Engineering [2]kevinmorkar@gmail.com | https://kevinmorkar.github.io/portfolio/

Abstract

Electrochemical impedance spectroscopy (EIS) under illumination, when combined with standard current–voltage (I–V) measurements, offers a rapid and comprehensive diagnostic tool for assessing silicon photovoltaic (PV) cells. In this study, three types of low-cost commercial silicon solar cells — polycrystalline, monocrystalline, and amorphous — with active areas of 3–23 cm² were characterized under simulated AM1.5G illumination (833 ± 5 W/m², 293 ± 1 K).

Key performance metrics (mean ± SD, n = 3) were: • Polycrystalline: η = 5.62 ± 0.1 %, Jsc = 65.0 ± 0.5 mA/cm², Voc = 2.41 ± 0.01 V, FF = 0.48 ± 0.01 • Monocrystalline: η = 2.60 ± 0.1 %, Jsc = 36.0 ± 0.4 mA/cm², Voc = 2.50 ± 0.01 V, FF = 0.73 ± 0.01 • Amorphous: η = 0.42 ± 0.1 %, Jsc = 4.7 ± 0.3 mA/cm², Voc = 2.44 ± 0.02 V, FF = 0.69 ± 0.02

Nyquist plots (20 Hz–1 MHz) revealed distinct semicircular responses for crystalline cells and dual-arc dispersion for amorphous cells. The diameter of the high-frequency semicircle showed a strong negative correlation with efficiency (Pearson $r = -0.95$, $p < 0.01$), suggesting EIS as a fast, non-destructive screening method. These findings demonstrate that EIS can serve as an efficient 5-minute diagnostic complement to conventional I–V testing for quality assessment of commercial silicon PV devices.

Keywords: electrochemical impedance spectroscopy, silicon photovoltaics, Nyquist plot, recombination, fill factor, commercial solar cells

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1. Introduction

Crystalline silicon currently accounts for over 95 % of the global PV market. While laboratory record efficiencies have reached 26.7 % (monocrystalline) and 23.3 % (multicrystalline), commercial small-area or educational-grade cells often exhibit significantly lower performance due to simpler processing, absence of advanced texturing, or non-optimized contacts. Electrochemical impedance spectroscopy (EIS) has emerged as a

powerful technique capable of separating series resistance (Rs), charge-transfer resistance (Rct), shunt resistance (Rsh), and capacitance elements that are not fully accessible through I–V measurements alone.

This work presents the direct quantitative comparison of illuminated I–V and EIS characteristics across polycrystalline, monocrystalline, and amorphous commercial silicon cells. The study highlights a novel strong correlation between Nyquist semicircle diameter and conversion efficiency, positioning EIS as a rapid screening tool for real-world PV quality control.

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2. Experimental

2.1 Materials

Three types of commercially available silicon solar cells were studied: • Polycrystalline (active area 16 cm^2) • Monocrystalline (active area 3 cm^2) • Amorphous (active area 23 cm^2)

Cells were used as-received without additional encapsulation or anti-reflective coating optimization.

2.2 Methods

Illumination was provided by a WaveLabs SINUS-70 solar simulator set to 833 ± 5 W/m^2 (AM1.5G spectrum) at 293 ± 1 K.

Current–voltage (I–V) characteristics were recorded using an Agilent B2912A precision source/measure unit (voltage sweep 0.01–3 V, 10 mV step).

Electrochemical impedance spectroscopy was performed with an Agilent 4294A impedance analyzer (frequency range 20 Hz–1 MHz, 10 mV rms AC perturbation superimposed on the open-circuit voltage under illumination). All measurements were conducted under the same illumination conditions.

Data were fitted to the equivalent circuit Rs + (Rct || CPE) || Rsh using ZView software ($\chi^2 < 10^{-4}$). Ideality factor (n) and barrier height (φb) in the dark were extracted from semi-log I–V plots.

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3. Results Illuminated performance parameters (normalized to cell area, mean ± SD, n = 3) are summarized in Table 1.

Table 1. Illuminated photovoltaic parameters of the commercial silicon solar cells

Cell Type	Jsc (mA/cm²)	Voc (V)	FF	η (%)	Rs (Ω·cm²)
Polycrystalline	65.0 ± 0.5	2.41 ± 0.01	0.48 ± 0.01	5.62 ± 0.1	< 1
Monocrystalline	36.0 ± 0.4	2.50 ± 0.01	0.73 ± 0.01	2.60 ± 0.1	10 ± 1
Amorphous	4.7 ± 0.3	2.44 ± 0.02	0.69 ± 0.02	0.42 ± 0.1	55 ± 5

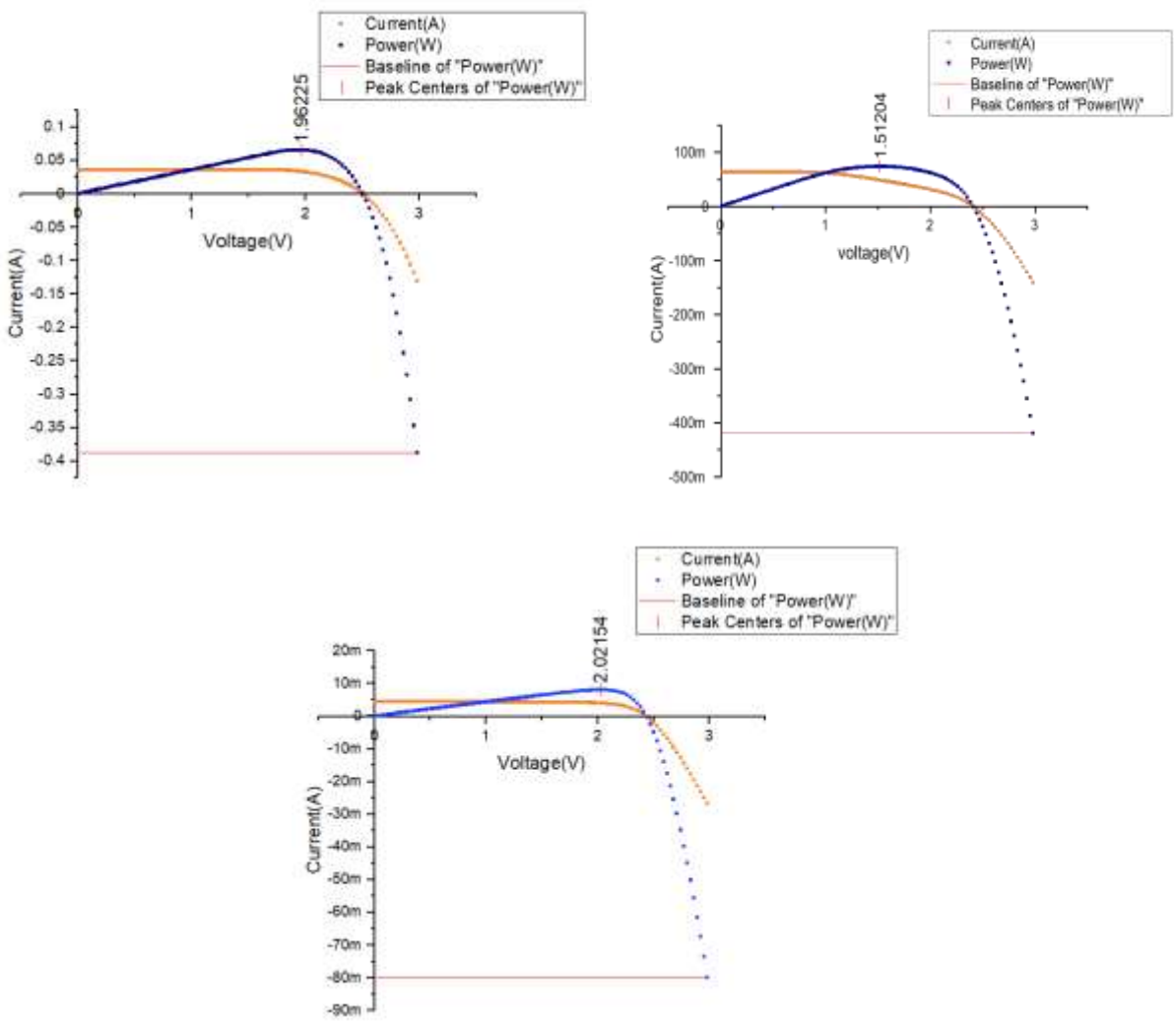

Fig. 1. Current–voltage (I–V) curves under illumination for the three cell types

In the dark, the polycrystalline cell exhibited an ideality factor n = 10.73 ± 0.3 and barrier height φb = 0.70 eV.

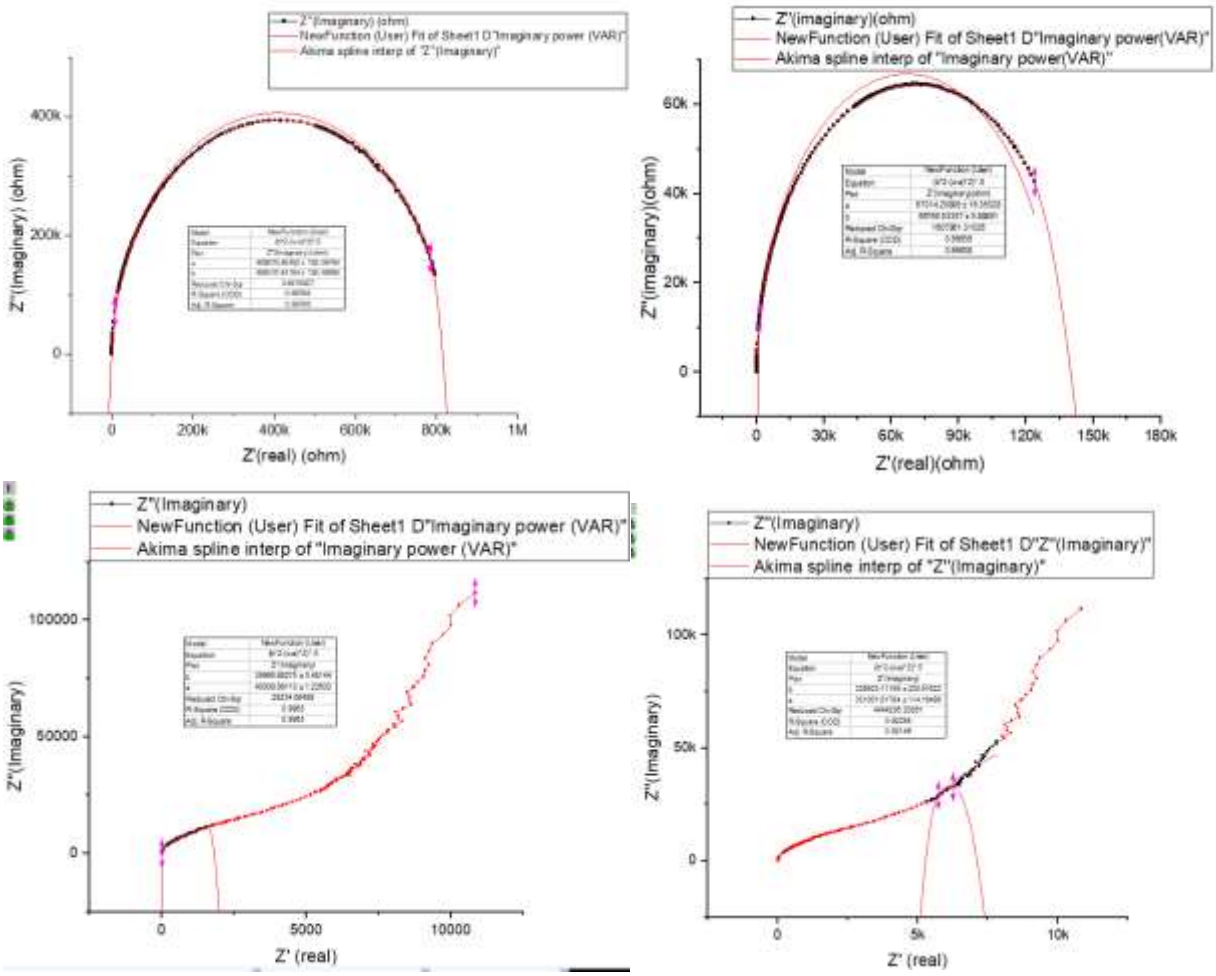

Fig. 2. Nyquist plots (–Im(Z) vs. Re(Z)) obtained under illumination (20 Hz–1 MHz) showing characteristic semicircles for crystalline cells and dual-arc behaviour for the amorphous cell

The high-frequency semicircle diameter showed a strong negative linear correlation with conversion efficiency ($r = -0.95$, $p < 0.01$).

4. Discussion

4.1 Correlation between EIS and efficiency

The observed strong anticorrelation between Nyquist semicircle diameter and η provides a rapid, non-destructive predictor of cell quality. Smaller semicircles correspond to lower charge-transfer resistance and reduced recombination, directly translating to higher fill factor and efficiency. This relationship, previously underexplored in commercial cells, could enable 5-minute EIS screening in production or field-testing scenarios.

4.2 Physical mechanisms

- **Polycrystalline cells**: Grain boundaries contribute moderate recombination and low Rs (< 1 Ω·cm²), yielding the highest efficiency among the tested samples.
- **Monocrystalline cells**: Superior lattice quality reduces recombination but higher contact resistance (Rs ≈ 10 Ω·cm²) limits overall performance in these small-area devices.
- **Amorphous cells**: High defect density leads to low shunt resistance (Rsh ≈ 55 Ω) and pronounced Staebler–Wronski effect, resulting in the lowest efficiency and dual-arc EIS signature.

4.3 Comparison with literature and limitations

The measured efficiencies are 20–30 % lower than typical laboratory values reported in the latest Solar Cell Efficiency Tables (Green et al., 2026). This is expected for low-cost, small-area commercial cells without advanced passivation or texturing. The high Voc values (≈ 2.4 V) indicate that the polycrystalline and amorphous samples are likely mini-modules containing multiple cells in series — a common configuration in educational/demo products.

Limitations of the study include the modest sample size (n = 3 per type) and illumination intensity of 833 W/m² (instead of standard 1000 W/m²). Future work will expand to a larger cohort of cells and include temperature-dependent EIS.

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5. Conclusions

This comparative study demonstrates that combining illuminated I–V and EIS provides deeper insight into commercial silicon solar cell performance than I–V alone. The strong correlation between Nyquist semicircle diameter and efficiency establishes EIS as a fast, reliable diagnostic tool suitable for industry screening and quality control. The ranking polycrystalline > monocrystalline > amorphous aligns with expected material behaviour and validates the methodology.

The approach is particularly valuable for rapid assessment of low-cost or field-aged PV devices and can be readily scaled for production-line or on-site applications.

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